How do RNNs encode mixtures? One network yields distinct geometries

Esther Poniatowski · Claire Sergent esther.poniatowski@ens.psl.eu
Integrative Neuroscience and Cognition Center (INCC), CNRS UMR 8002, Université Paris Cité, Paris, France

SCAN FOR
supporting details,
references, contacts
esther-poniatowski.github.io/royaumont2026

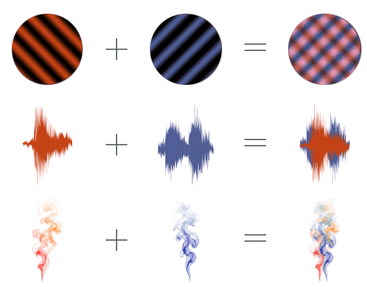
1 PROBLEM
Natural inputs often mix simpler components

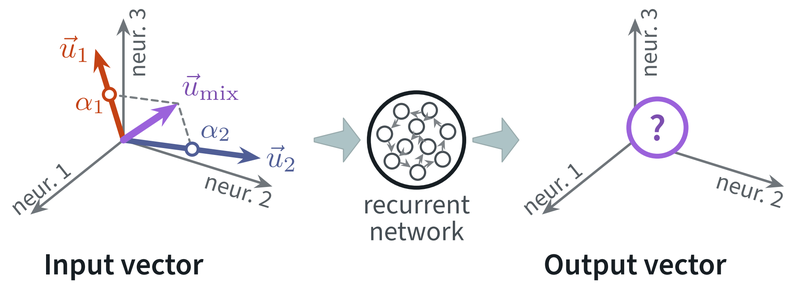
Weighted sum of two components: $\vec{u}_{\text{mix}} = \alpha_1 \vec{u}_1 + \alpha_2 \vec{u}_2$

Vision
overlapping gratings

Audition
mixed voices

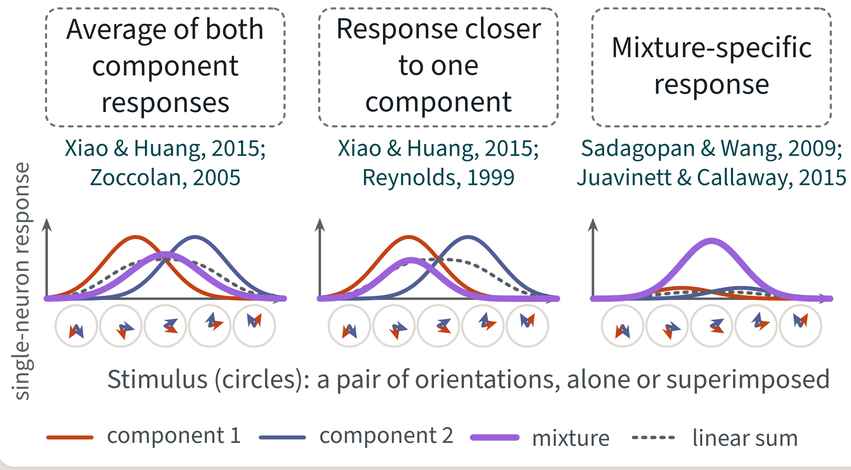
Olfaction
odour blends





How does a whole network encode a mixture of inputs?

2 OBSERVATION
Single neurons respond heterogeneously

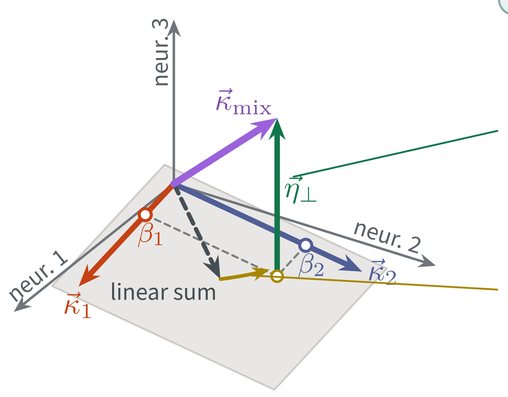


3 DESCRIPTION FRAMEWORK
The mixture geometry is compared to the components'

Reference: the outputs to each component alone, $\vec{\kappa}_1$ and $\vec{\kappa}_2$, and their linear sum $\alpha_1\vec{\kappa}_1 + \alpha_2\vec{\kappa}_2$.

$$\vec{\kappa}_{\text{mix}} = \underbrace{\beta_1\vec{\kappa}_1 + \beta_2\vec{\kappa}_2}_{\text{in the components plane}} + \underbrace{\beta_\perp\vec{\eta}_\perp}_{\text{orthogonal to the plane}}$$

TWO POTENTIAL DEVIATIONS
from the linear sum



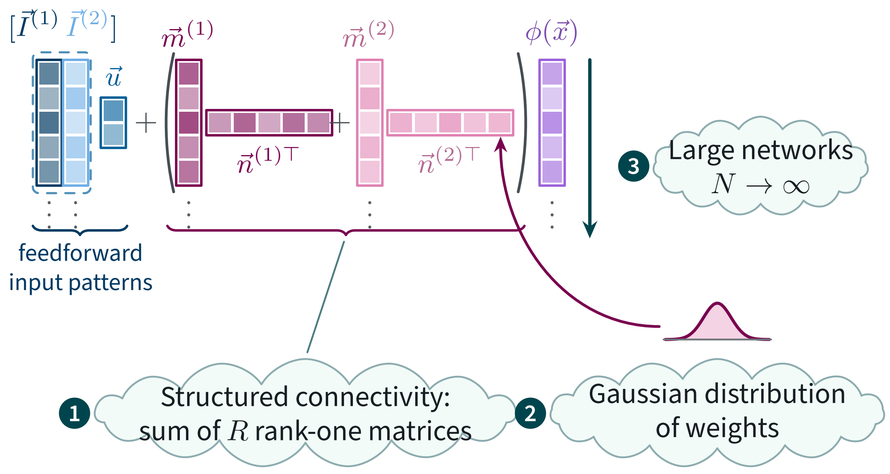
EMERGENCE
out of the components plane
$\beta_\perp \neq 0$

BIAS
relative to the components weights
$\beta_1{:}\beta_2 \neq \alpha_1{:}\alpha_2$

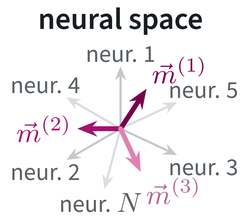
4 MODELING APPROACH
A reduced model links network structure to geometry

A few biologically plausible assumptions reduce a standard rate model to a set of collective statistics (Mastrogiuseppe & Ostojic, 2018; Beiran et al., 2021; Dubreuil et al., 2022).

$$\tau\,\dot{\vec{x}} = -\vec{x} + \left[\,\vec{I}^{(1)}\ \vec{I}^{(2)}\,\right]\vec{u} + \frac{1}{N}\sum_{r=1}^{R}\vec{m}^{(r)}\vec{n}^{(r)\top}\phi(\vec{x})$$

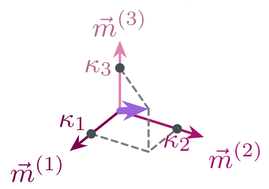


High-dimensional neural space



mean-field reduction

Reduced coordinates capture output geometry



5 MECHANISM
Two network operations determine the geometry

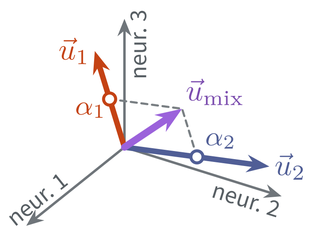
Both operations are explicit along the principal directions of activity fixed by connectivity.

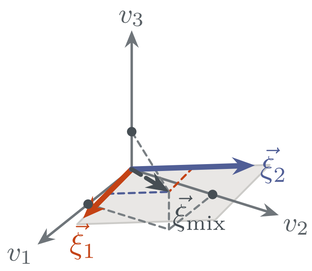
FEEDFORWARD PROJECTION

maps inputs onto population patterns
↓
linear combination of the components

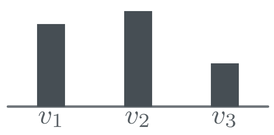
$$\vec{\xi}_{\text{mix}} = \alpha_1\vec{\xi}_1 + \alpha_2\vec{\xi}_2$$

input space

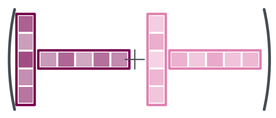




coordinates of the projection along the principal directions

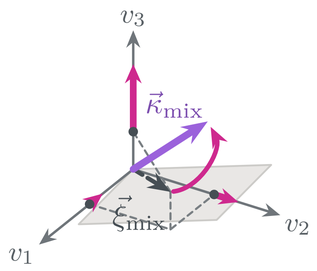


RECURRENT REWEIGHTING

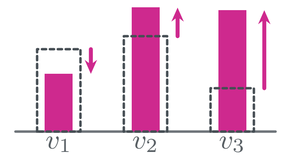


modulates the contribution of each elicited pattern
↓
rotation relative to the linear sum

$$\vec{\kappa}_{\text{mix}} = A_1\,\xi_{\text{mix},1}\,\vec{v}_1 + A_2\,\xi_{\text{mix},2}\,\vec{v}_2 + A_3\,\xi_{\text{mix},3}\,\vec{v}_3$$



coordinates after reweighting

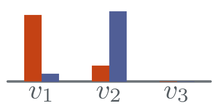


Bars: coordinates along the principal directions v_k
component 1 · component 2
projected mixture · after reweighting

FEEDFORWARD PROJECTION

How many eigendir. do the components activate jointly?

2 active eigendir.



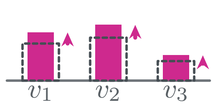
≥ 3 active eigendir.



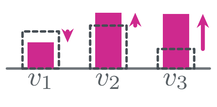
RECURRENT REWEIGHTING

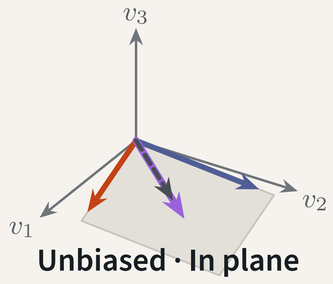
How is the mixture's projection rescaled across the active eigendirections?

uniform scaling

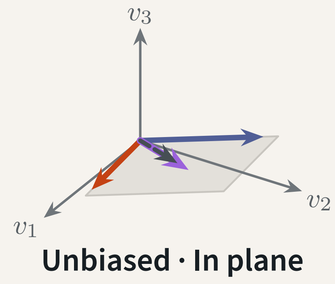


uneven scaling

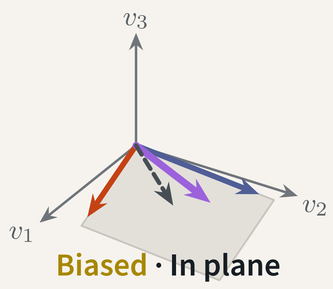




Unbiased · In plane
output aligned with the linear sum

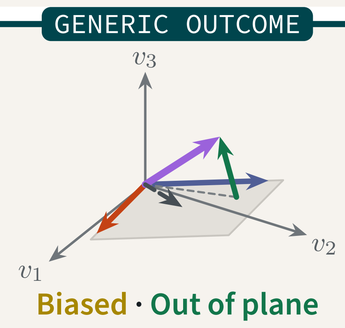


Unbiased · In plane
output aligned with the linear sum



Biased · In plane
output rotates within the plane

GENERIC OUTCOME



Biased · Out of plane
output rotates out of the plane

6 INTERPRETATIONS
Each geometric property shapes what a readout can infer

- A **biased** combination can implement attentional selection by **gain control**.
- An **emergent** pattern is **linearly separable** from its components, readable as a distinct identity.

7 PREDICTIONS
The model predicts testable signatures

- **Emergence requires dissimilar components**, and grows as they separate.
- **Bias decreases** with stimulus intensity.

Royaumont Workshop 2026 · Abbaye de Royaumont
Global brain dynamics and the shaping of mental representations

Composition and colour advice · Krees Sama